

Homework #6

Due: 11:59pm, March 13

Objectives: Study filter design methods**1. [Generalized linear phase]**

Consider an FIR filter $h[n]$ of length L with real-valued coefficients. Suppose that L is an odd number. Furthermore, we assume the anti-symmetry of the filter coefficients so that

$$h[n] = -h[L - 1 - n].$$

- (a) Show that the “continuous-phase” $\phi(\omega)$ is written in the form of $\omega\ell + \phi_0$ for $2\ell \in \mathbb{Z}$ and $\phi_0 \in \{0, \frac{\pi}{2}\}$. Specify ℓ and ϕ_0 .
- (b) Show that $h[n]$ has zero DC gain, that is,

$$H(e^{j0}) = \sum_{n=0}^{L-1} h[n] = 0.$$

- (c) Show that $h[n]$ has zero HF gain, that is,

$$H(e^{j\pi}) = \sum_{n=0}^{L-1} h[n](-1)^n = 0.$$

2. [OS 7.31]

A discrete-time highpass filter can be obtained from a continuous-time lowpass filter by the following transformation:

$$H(z) = H_c(s)|_{s=(1+z^{-1})/(1-z^{-1})}.$$

- (a) Show that this transformation maps the $j\Omega$ -axis of the s -plane onto the unit circle of the z -plane.
- (b) Show that if $H_c(s)$ is a rational function with all its poles inside the left-half s -plane, then $H(z)$ will be a rational function with all its poles inside the unit circle of the z -plane.
- (c) Suppose a desired highpass discrete-time filter has specifications

$$\begin{aligned} |H(e^{j\omega})| &\leq 0.01, & |\omega| &\leq \pi/3 \\ 0.95 &\leq |H(e^{j\omega})| \leq 1.05, & \pi/2 &\leq |\omega| \leq \pi. \end{aligned}$$

Determine the specification on the continuous-time lowpass filter so that the desired highpass discrete-time results from the transformation specified at the beginning of this problem.

3. **[OS 7.30, 3rd edition]**

Consider designing a discrete-time filter with system function $H(z)$ from a continuous-time filter with rational system function $H_c(s)$ by the transformation

$$H(z) = H_c(s) \Big|_{s=\beta[(1-z^{-\alpha})/(1+z^{-\alpha})]},$$

where α is a nonzero integer and β is a real number.

- (a) If $\alpha > 0$, for what values of β does a stable, causal continuous-time filter with rational $H_c(s)$ always lead to a stable, causal discrete-time filter with rational $H(z)$?
- (b) If $\alpha < 0$, for what values of β does a stable, causal continuous-time filter with rational $H_c(s)$ always lead to a stable, causal discrete-time filter with rational $H(z)$?
- (c) For $\alpha = 2$ and $\beta = 1$, determine to what contour in the z -plane the $j\Omega$ -axis of the s -plane maps.
- (d) Suppose that the continuous-time filter is a stable lowpass filter with passband frequency response such that

$$1 - \delta \leq |H_c(j\Omega)| \leq 1 + \delta, \quad |\Omega| \leq 1.$$

If the discrete-time system $H(z)$ is obtained by the transformation set forth at the beginning of this problem, with $\alpha = 2$ and $\beta = 1$, determine the value of ω in the interval $[-\pi, \pi)$, for which

$$1 - \delta \leq |H(e^{j\omega})| \leq 1 + \delta.$$

4. **[IIR filter design in the time domain]**

We would like to design an IIR lowpass filter with the following specifications

$$\omega_p = 0.2\pi, \quad \omega_s = 0.3\pi, \quad \delta_p \leq 0.02, \quad \delta_s \leq 0.01.$$

You may use the scripts in the following address: https://johnloomis.org/ece562/hayes_book/model_m.html.

- (a) Design an IIR filter by Butterworth design (use `buttord` and `butter`). Compute its impulse response by `impz` and plot it. Let us consider a truncated impulse response consisting of the first 100 samples.
- (b) Design an IIR filter by approximating Butterworth design in part (a) with Padé's method. Find N and M (smaller than the order of Butterworth) by trial and error while satisfying the given specifications. Repeat this for Prony's method and for Shank's method.
- (c) Repeat part (b) for Prony's method.
- (d) Repeat part (b) for Shank's method.